

Local Results Summary — Consolidated + Blocked Models

Two-page summary: page 1 is the consolidated “All values, row-by-row” reference covering every (model × benchmark × mode) tuple we have on disk, plus the hook-aware best-across-α-triples columns. Page 2 is the “Blocked models” explanation listing the per-model infrastructure blockers (handoffs filed / in-session patches applied / structural blocks). For the per-model BRV-comparison tables, the closeness analysis, observations, and provenance, see the full PDF local_results_uniform_lambda_0.9.pdf.

All values, row-by-row (consolidated reference)

Coverage: 8 of the 9 enrichment-target VLMs now appear in this table (LLaVA-1.5-7B, Idefics2-8B, LLaVA-Next-LLaMA3-8B, LLaVA-Next-Mistral-7B, Qwen2-VL-7B-Instruct, LLaVA-OneVision-7B, InternVL2.5-8B, Qwen2.5-VL-7B). The 3 newly-added models have partial cells (uniform-only or baseline-only) because their eval jobs are still in flight or only one side has landed so far. Qwen2.5-VL-3B remains structurally blocked (no shape-compatible math donor); see the “Blocked models” page for the full blocker map.

Note on (NEW) rows for LLaVA-1.5 and Mistral: the MathVista-General, MathVista-Math, and DynaMath-Worst values for those two pairs are computed by extract_merge_coverage.py from per-skill MathVista CSV rows / row 3 of the DynaMath CSV. MathVista-Math and DynaMath-Worst are exact matches against the canonical extraction. MathVista-General is approximated as (Overall – Math) weighted by row count; BRV’s General excludes additional math-leaning skills (arithmetic, algebra, geometry, etc.) so its published value tends to run ~5 pp higher than this approximation. The 3 existing models (LLaVA-LLaMA3, Idefics2, Qwen2-VL) keep the original BRV-script values.

Same data as the per-model tables above, flattened into one table for spreadsheet-style scanning. One row per (model × benchmark). **n (Local)** is the row count we measure in the local ~/LMUData/ TSV; **n (BRV)** is the row count BRV's eval registers via VLMEvalKit_brv/vlmeval/dataset/image_vqa.py. The two columns are identical for every row because BRV's dataset registry resolves to the same upstream files (MD5-matched). The asterisk on MathVerse-Overall (788*) marks the *effective* n — the five modes share the same 788 problems, so the 5-mode average denominator stays at 788, not 3,940. Red highlight on Gap cells = |Local – Paper| > 1 pp.

					Baseline			+Uniform 0.9			Δ	
Model	Benchmark	n (Local)	n (BRV)	Type	Local	Paper	Gap	Local	Paper	Gap	Local	Paper
LLaVA-Next-LLaMA3-8B (Dart-Prop)	MathVista-Overall	1000	1000	perception	37.80	37.40	+0.40	37.90	38.00	-0.10	+0.10	+0.60
	MathVista-General (NEW)	460	460	perception	51.96	51.70	+0.26	49.13	48.70	+0.43	-2.83	-3.00
	MathVista-Math (NEW)	540	540	reasoning	25.74	25.40	+0.34	28.33	28.90	-0.57	+2.59	+3.50
	MathVerse-Overall	788*	788*	reasoning	18.45	20.10	-1.65	19.03	23.70	-4.67	+0.58	+3.60
	MathVerse-T-D	788	788	reasoning	22.34	25.90	-3.56	23.98	30.70	-6.72	+1.64	+4.80
	MathVerse-T-L	788	788	reasoning	19.67	20.80	-1.13	19.92	24.80	-4.88	+0.25	+4.00
	MathVerse-V-I	788	788	reasoning	19.80	21.10	-1.30	20.18	25.50	-5.32	+0.38	+4.40
	MathVerse-V-D	788	788	reasoning	15.48	16.50	-1.02	15.48	19.80	-4.32	+0.00	+3.30
	MathVerse-V-O	788	788	reasoning	14.97	16.00	-1.03	15.61	17.40	-1.79	+0.64	+1.40
	MathVision	304	304	reasoning	14.14	13.80	+0.34	14.47	14.80	-0.33	+0.33	+1.00
	MMStar-Overall	1500	1500	perception	43.33	43.80	-0.47	43.60	43.60	+0.00	+0.27	-0.20
	MMStar-Math	250	250	reasoning	29.60	30.00	-0.40	33.60	33.60	+0.00	+4.00	+3.60
	DynaMath-Average	5010	5010	reasoning	22.75	22.70	+0.05	24.35	24.50	-0.15	+1.60	+1.80
	DynaMath-Worst (NEW)	501	501	reasoning	3.19	—	—	4.39	—	—	+1.20	—
Idefics2-8B (MAmmoTH-1)	MathVista-Overall	1000	1000	perception	52.00	51.80	+0.20	52.50	53.00	-0.50	+0.50	+1.20
	MathVista-General (NEW)	460	460	perception	56.30	57.00	-0.70	59.13	58.50	+0.63	+2.83	+1.50
	MathVista-Math (NEW)	540	540	reasoning	48.33	47.40	+0.93	46.85	48.30	-1.45	-1.48	+0.90
	MathVerse-Overall	788*	788*	reasoning	18.38	19.40	-1.02	18.96	20.40	-1.44	+0.58	+1.00
	MathVerse-T-D	788	788	reasoning	21.70	24.40	-2.70	24.62	26.00	-1.38	+2.92	+1.60
	MathVerse-T-L	788	788	reasoning	20.81	21.30	-0.49	21.57	22.50	-0.93	+0.76	+1.20
	MathVerse-V-I	788	788	reasoning	20.05	20.70	-0.65	19.67	21.30	-1.63	-0.38	+0.60
	MathVerse-V-D	788	788	reasoning	19.04	19.70	-0.66	17.89	19.80	-1.91	-1.15	+0.10
	MathVerse-V-O	788	788	reasoning	10.28	11.00	-0.72	11.04	12.10	-1.06	+0.76	+1.10
	MathVision	304	304	reasoning	16.45	17.10	-0.65	14.80	16.80	-2.00	-1.65	-0.30
	MMStar-Overall	1500	1500	perception	49.33	49.50	-0.17	48.40	48.30	+0.10	-0.93	-1.20

					Baseline			+Uniform 0.9			Δ	
Model	Benchmark	n (Local)	n (BRV)	Type	Local	Paper	Gap	Local	Paper	Gap	Local	Paper
	MMStar-Math	250	250	reasoning	39.60	39.60	+0.00	40.00	40.80	-0.80	+0.40	+1.20
	DynaMath-Average	5010	5010	reasoning	21.80	21.80	+0.00	23.01	23.20	-0.19	+1.21	+1.40
	DynaMath-Worst (NEW)	501	501	reasoning	3.19	—	—	4.79	—	—	+1.60	—
Qwen2-VL-7B-Instruct (Qwen2-Math-7B)	MathVista-Overall	1000	1000	perception	61.10	61.20	-0.10	58.30	60.20	-1.90	-2.80	-1.00
	MathVista-General (NEW)	460	460	perception	69.78	69.60	+0.18	66.09	68.00	-1.91	-3.69	-1.60
	MathVista-Math (NEW)	540	540	reasoning	53.70	54.10	-0.40	51.67	53.50	-1.83	-2.03	-0.60
	MathVerse-Overall	788*	788*	reasoning	30.68	31.80	-1.12	30.41	31.90	-1.49	-0.27	+0.10
	MathVerse-T-D	788	788	reasoning	34.77	35.90	-1.13	33.88	37.10	-3.22	-0.89	+1.20
	MathVerse-T-L	788	788	reasoning	30.58	31.40	-0.82	30.08	31.70	-1.62	-0.50	+0.30
	MathVerse-V-I	788	788	reasoning	30.71	31.50	-0.79	30.08	31.50	-1.42	-0.63	+0.00
	MathVerse-V-D	788	788	reasoning	32.23	33.10	-0.87	31.09	32.50	-1.41	-1.14	-0.60
	MathVerse-V-O	788	788	reasoning	25.13	26.90	-1.77	26.90	26.70	+0.20	+1.77	-0.20
	MathVision	304	304	reasoning	20.72	21.10	-0.38	20.07	21.70	-1.63	-0.65	+0.60
	MMStar-Overall	1500	1500	perception	59.60	59.90	-0.30	56.47	59.50	-3.03	-3.13	-0.40
	MMStar-Math	250	250	reasoning	58.40	59.20	-0.80	55.20	58.40	-3.20	-3.20	-0.80
	DynaMath-Average	5010	5010	reasoning	33.55	34.40	-0.85	31.40	35.00	-3.60	-2.15	+0.60
	DynaMath-Worst (NEW)	501	501	reasoning	10.38	—	—	10.78	—	—	+0.40	—
LLaVA-1.5-7B (MetaMath-7B-V1.0)	MathVista-Overall	1000	1000	perception	24.10	—	—	23.70	—	—	-0.40	—
	MathVista-General (NEW)	460	460	perception	24.21	—	—	23.63	—	—	-0.58	—
	MathVista-Math (NEW)	540	540	reasoning	17.04	—	—	18.33	—	—	+1.29	—
	MathVerse-Overall	788*	788*	reasoning	15.03	—	—	14.90	—	—	-0.13	—
	MathVerse-T-D	788	788	reasoning	15.61	—	—	14.85	—	—	-0.76	—
	MathVerse-T-L	788	788	reasoning	14.59	—	—	14.59	—	—	+0.00	—
	MathVerse-V-I	788	788	reasoning	14.85	—	—	14.72	—	—	-0.13	—
	MathVerse-V-D	788	788	reasoning	15.36	—	—	15.10	—	—	-0.26	—
	MathVerse-V-O	788	788	reasoning	14.72	—	—	15.23	—	—	+0.51	—
	MathVision	304	304	reasoning	15.79	—	—	17.11	—	—	+1.32	—
	MMStar-Overall	1500	1500	perception	31.07	—	—	—	—	—	—	—
	MMStar-Math	250	250	reasoning	22.40	—	—	—	—	—	—	—
	DynaMath-Average	5010	5010	reasoning	12.73	—	—	11.86	—	—	-0.87	—
	DynaMath-Worst (NEW)	501	501	reasoning	1.20	—	—	1.80	—	—	+0.60	—
LLaVA-Next-Mistral-7B (MAmmoTH-7B-Mistral)	MathVista-Overall	1000	1000	perception	35.40	—	—	36.00	—	—	+0.60	—
	MathVista-General (NEW)	460	460	perception	35.96	—	—	36.74	—	—	+0.78	—
	MathVista-Math (NEW)	540	540	reasoning	26.11	—	—	26.30	—	—	+0.19	—
	MathVerse-Overall	788*	788*	reasoning	18.27	—	—	17.92	—	—	-0.35	—
	MathVerse-T-D	788	788	reasoning	21.19	—	—	21.07	—	—	-0.12	—

					Baseline			+Uniform 0.9			Δ	
Model	Benchmark	n (Local)	n (BRV)	Type	Local	Paper	Gap	Local	Paper	Gap	Local	Paper
	MathVerse-T-L	788	788	reasoning	17.89	—	—	18.27	—	—	+0.38	—
	MathVerse-V-I	788	788	reasoning	19.16	—	—	19.16	—	—	+0.00	—
	MathVerse-V-D	788	788	reasoning	17.01	—	—	16.75	—	—	-0.26	—
	MathVerse-V-O	788	788	reasoning	16.12	—	—	14.34	—	—	-1.78	—
	MathVision	304	304	reasoning	11.18	—	—	13.82	—	—	+2.64	—
	MMStar-Overall	1500	1500	perception	39.33	—	—	—	—	—	—	—
	MMStar-Math	250	250	reasoning	27.20	—	—	—	—	—	—	—
	DynaMath-Average	5010	5010	reasoning	17.78	—	—	19.26	—	—	+1.48	—
	DynaMath-Worst (NEW)	501	501	reasoning	2.20	—	—	2.40	—	—	+0.20	—
LLaVA-OneVision-7B (Qwen2-Math-7B)	MathVista-Overall	1000	1000	perception	—	—	—	60.00	—	—	—	—
	MathVista-General (NEW)	460	460	perception	—	—	—	59.31	—	—	—	—
	MathVista-Math (NEW)	540	540	reasoning	—	—	—	61.48	—	—	—	—
	MathVerse-Overall	788*	788*	reasoning	—	—	—	30.36	—	—	—	—
	MathVerse-T-D	788	788	reasoning	—	—	—	38.71	—	—	—	—
	MathVerse-T-L	788	788	reasoning	—	—	—	33.63	—	—	—	—
	MathVerse-V-I	788	788	reasoning	—	—	—	31.73	—	—	—	—
	MathVerse-V-D	788	788	reasoning	—	—	—	29.06	—	—	—	—
	MathVerse-V-O	788	788	reasoning	17.64	—	—	18.65	—	—	+1.01	—
	MathVision	304	304	reasoning	18.42	—	—	17.43	—	—	-0.99	—
	MMStar-Overall	1500	1500	perception	0.00	—	—	0.00	—	—	+0.00	—
	MMStar-Math	250	250	reasoning	0.00	—	—	0.00	—	—	+0.00	—
	DynaMath-Average	5010	5010	reasoning	19.66	—	—	20.72	—	—	+1.06	—
	DynaMath-Worst (NEW)	501	501	reasoning	4.99	—	—	5.59	—	—	+0.60	—
InternVL2.5-8B (InternLM2-Math-Plus-7B)	MathVista-Overall	1000	1000	perception	65.00	—	—	—	—	—	—	—
	MathVista-General (NEW)	460	460	perception	64.59	—	—	—	—	—	—	—
	MathVista-Math (NEW)	540	540	reasoning	61.48	—	—	—	—	—	—	—
	MathVerse-Overall	788*	788*	reasoning	34.67	—	—	27.36	—	—	-7.31	—
	MathVerse-T-D	788	788	reasoning	42.89	—	—	31.60	—	—	-11.29	—
	MathVerse-T-L	788	788	reasoning	38.58	—	—	29.06	—	—	-9.52	—
	MathVerse-V-I	788	788	reasoning	37.31	—	—	30.20	—	—	-7.11	—
	MathVerse-V-D	788	788	reasoning	33.25	—	—	26.14	—	—	-7.11	—
	MathVerse-V-O	788	788	reasoning	21.32	—	—	19.80	—	—	-1.52	—
	MathVision	304	304	reasoning	23.68	—	—	10.86	—	—	-12.82	—
	MMStar-Overall	1500	1500	perception	62.73	—	—	61.13	—	—	-1.60	—
	MMStar-Math	250	250	reasoning	68.00	—	—	69.60	—	—	+1.60	—
	DynaMath-Average	5010	5010	reasoning	32.22	—	—	—	—	—	—	—
	DynaMath-Worst (NEW)	501	501	reasoning	9.38	—	—	—	—	—	—	—

					Baseline			+Uniform 0.9			Δ	
Model	Benchmark	n (Local)	n (BRV)	Type	Local	Paper	Gap	Local	Paper	Gap	Local	Paper
Qwen2.5-VL-7B (Qwen2.5-Math-7B)	MathVista-Overall	1000	1000	perception	68.90	—	—	—	—	—	—	—
	MathVista-General (NEW)	460	460	perception	68.20	—	—	—	—	—	—	—
	MathVista-Math (NEW)	540	540	reasoning	67.04	—	—	—	—	—	—	—
	MathVerse-Overall	788*	788*	reasoning	44.01	—	—	—	—	—	—	—
	MathVerse-T-D	788	788	reasoning	53.43	—	—	—	—	—	—	—
	MathVerse-T-L	788	788	reasoning	45.18	—	—	—	—	—	—	—
	MathVerse-V-I	788	788	reasoning	40.61	—	—	—	—	—	—	—
	MathVerse-V-D	788	788	reasoning	40.10	—	—	—	—	—	—	—
	MathVerse-V-O	788	788	reasoning	40.74	—	—	—	—	—	—	—
	MathVision	304	304	reasoning	26.32	—	—	—	—	—	—	—
	MMStar-Overall	1500	1500	perception	61.67	—	—	—	—	—	—	—
	MMStar-Math	250	250	reasoning	61.20	—	—	—	—	—	—	—
	DynaMath-Average	5010	5010	reasoning	—	—	—	—	—	—	—	—
	DynaMath-Worst (NEW)	501	501	reasoning	—	—	—	—	—	—	—	—

Blocked models — no on-disk data

These 4 models complete the 9-model enrichment-target set but have no step-25 score CSVs at all. Each is blocked by a different infrastructure issue, summarized here.

Model	Auto-routed math LLM	Blocker	Status / handoff
LLaVA-OneVision-7B	Qwen2-Math-7B (HF)	VLMEvalKit_brv venv missing the <i>//lava</i> Python package. Eval jobs die silently with No module named 'llava'. Same package issue affects LLaVA-1.5-7B today — both will resume once the package is installed. Merge .pth already exists from earlier session.	Handoff filed: notes/vlmevalkit_handoff_llava_next_install.md. One uv pip install command unblocks both llava-1.5 and llava-ov.
Qwen2.5-VL-7B	Qwen2.5-Math-7B (HF)	Two blockers. (1) Merge job crashed twice with an hf-xet Rust panic mid-download of Qwen2.5-Math-7B (range end out of bounds: 67M ≤ 34M). Worked around by pre-fetching via huggingface-cli download; merge re-submitted 2026-06-04 18:32. (2) Every eval job raises KeyError: 'Qwen2.5-VL-7B-Instruct' — VLMEvalKit's supported_VLM registry has the 3B and 2B but not the 7B variant.	Handoff filed: notes/vlmevalkit_handoff_qwen25vl_7b_registry.md (eval). Merge unblock used huggingface-cli download Qwen/Qwen2.5-Math-7B to bypass hf-xet.
InternVL2.5-8B	internlm2-math-plus-7b (HF)	Merge job formerly killed by TERM_THREADLIMIT (fixed in run_pipeline.sh) and raised TypeError: dtype in merge_pmbt.py (fixed in code/merge_pmbt.py). Merge .pth landed 2026-06-04 14:56 (15.5 GB) . Eval still fails with 'InternLM2ForCausalLM' object has no attribute 'generate' — HF-cache modeling_internlm2.py doesn't inherit GenerationMixin.	Handoff filed: notes/internvl_handoff_generation_mixin.md. Merge side is unblocked by today's commits; eval side still needs the HF-cache patch (or transformers ≤ 4.49 pin).
Qwen2.5-VL-3B	— (none compatible)	Hard-blocked by run_pipeline.sh (exit 1). Qwen2.5-VL-3B uses a 36-layer 2048-hidden Qwen2.5-3B-base LLM; Qwen2.5-Math ships only 1.5B (28L/1536H) and 7B (28L/3584H), so no Qwen2.5-Math variant is shape-compatible. Other math families (LLaMA-base, Mistral-base) are also architecturally incompatible with the Qwen2.5-3B backbone.	Decision: leave the block in place . No shape-compatible math donor exists; lifting the block would just move the failure into merge_pmbt.py's tensor-shape mismatch path.